

APPLIED MACHINE LEARNING FOR TEXT ANALYTICS (24ALT3101)

REVIEW – 1

Title

Emotion Recognition from Social Media Text Using Large Language Models

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Introduction

Why Emotion Recognition?

- Social media platforms generate massive volumes of text daily, where people express thoughts, opinions, and emotions.
- Understanding these emotions helps organizations, businesses, and researchers analyze public opinion, improve customer satisfaction, and monitor mental well-being.
- Traditional sentiment analysis only classifies text as positive, negative, or neutral — it cannot capture specific emotions like joy, sadness, anger, fear, surprise, or love.
- This project proposes an Emotion Recognition system using Large Language Models (LLMs) to automatically detect fine-grained emotions from social media posts.
- The system fine-tunes Transformer-based models such as BERT, DistilBERT, or RoBERTa on benchmark emotion datasets to move beyond simple polarity detection.

Objectives

What the System sets out to do

1. Detect fine-grained emotions — classify posts into Joy, Sadness, Anger, Fear, Surprise, Love (and Neutral) instead of a simple positive/negative/neutral score.
2. Fine-tune pretrained LLMs — adapt Transformer encoders (BERT, DistilBERT, RoBERTa) for the emotion classification task.
3. Build a complete NLP pipeline — covering preprocessing, tokenization, normalization, feature extraction, training, and evaluation.
4. Evaluate rigorously — using accuracy, precision, recall, F1-score, and confusion matrix.
5. Deliver real-time predictions — through a web interface that returns the predicted emotion with a confidence score.

Literature Survey

- **Traditional Sentiment Analysis:** Early approaches used lexicon-based methods and classical ML (Naive Bayes, SVM) with bag-of-words/TF-IDF to classify text as positive/negative/neutral. They struggled with context, sarcasm, and nuanced emotion.
- **Deep Learning (RNN/LSTM/CNN):** Improved on classical ML by capturing sequential patterns, but still couldn't model long-range dependencies well.
- **Transformer Models (BERT, 2018):** Enabled deep bidirectional context understanding, greatly improving performance on emotion classification tasks — the foundation of modern approaches. Variants like DistilBERT (lighter, faster) and RoBERTa (more robust, higher accuracy) further improved on this
- **GoEmotions Dataset (2020):** A large, human-annotated dataset with 27 fine-grained emotion categories, enabling research to move beyond coarse positive/negative/neutral sentiment.
- **Research Gap:** Most existing systems still only detect coarse sentiment. This project addresses that gap by fine-tuning LLMs on emotion-labeled datasets to detect specific emotions in social media text.

Methodology

Approach – Data to Feature

- **Data Collection:** Gather social media text from the GoEmotions dataset and/or the Kaggle Emotion Dataset.
- **Data Preprocessing:** Remove URLs, symbols, and special characters; convert text to lowercase; perform tokenization, stop-word removal, stemming, and lemmatization.
- **Feature Extraction:** Convert pre-processed text into numerical representations using Transformer tokenizers.
- **Model Training:** Fine-tune a pretrained LLM (BERT, DistilBERT, or RoBERTa) for multi-class emotion classification.
- **Emotion Classification:** Predict emotions — Joy, Sadness, Anger, Fear, Surprise, Love, Neutral — from input text, along with a confidence score.
- **Model Evaluation:** Assess performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- **Result Visualization:** Display predicted emotion with confidence score; visualize emotion distribution using charts/graphs.

Conclusion

A Smarter Way to Understand Emotion

- This project demonstrates that fine-tuning Transformer-based LLMs (BERT, DistilBERT, RoBERTa) enables accurate, fine-grained emotion recognition from social media text, going beyond traditional positive/negative/neutral sentiment analysis.
- The system offers practical value across social media monitoring, customer feedback analysis, mental health awareness, public opinion analysis, and business decision-making.
- Future work could explore multilingual emotion detection, handling of sarcasm/mixed emotions, and lighter models for deployment on edge devices.

THANK YOU